PROGRAM:

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#include <stdio.h>
#include <stdlib.h>
 #define MAX 100 // Maximum size of the stack
// Stack structure definition
struct Stack {
    int arr[MAX];
    int top;
 };
// Function prototypes
void initializeStack(struct Stack *stack);
void push(struct Stack *stack, int element);
 int pop(struct Stack *stack);
 void displayStack(struct Stack *stack);
 int main() {
    struct Stack stack;
    int choice, element;
    initializeStack(&stack);
    while (1) {
        printf("\nStack Operations Menu:\n");
        printf("1. Push\n");
        printf("2. Pop\n");
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    printf("3. Display\n");
    printf("4. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            printf("Enter element to push: ");
            scanf("%d", &element);
            push(&stack, element);
            break;
        case 2:
            pop(&stack);
break;
        case 3:
            displayStack(&stack);
            break;
        case 4:
            exit(0);
        default:
            printf("Invalid choice! Please enter a valid option.\n");
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        }
    }
    return 0;
}
// Function to initialize the stack
void initializeStack(struct Stack *stack) {
    stack->top = -1;
}
// Function to push an element onto the stack
void push(struct Stack *stack, int element) {

    if (stack->top == (MAX-1)) {
        printf("Error: Stack overflow. Cannot push element.\n");
        return;
    }
    stack->arr[++stack->top] = element;
    printf("Element pushed: %d\n", element);
}
// Function to pop an element from the stack
int pop(struct Stack *stack) {
    if (stack->top == -1) {
        printf("Error: Stack underflow. Cannot pop element.\n");
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    }
    else  {
     printf(“\nPopped element: %d\n”, stack->arr[stack->top--]);
 }
// Function to display the stack elements
void displayStack(struct Stack *stack) {
    if (stack->top == -1) {
        printf("Stack is empty.\n");
        return;
   }
    printf("Stack elements: ");
    for (int i = stack->top; i >= 0; i--) {
       printf("%d ", stack->arr[i]);
    }
    printf("\n");

}
//OUTPUT:
Stack Operations Menu:
1. Push
2. Pop
```

3. Display

4. Exit Enter your choice: 1

Enter element to push: 10

Element pushed: 10

Stack Operations Menu:

1. Push

2. Pop

3. Display

4. Exit Enter your choice: 1

Enter element to push: 20

Element pushed: 20

Stack Operations Menu:

1. Push

2. Pop

3. Display

4. Exit

Enter your choice: 3

Stack elements: 20 10

Stack Operations Menu:

1. Push

2. Pop

3. Display

4. Exit

Enter your choice: 2

Popped element: 20

Stack Operations Menu:

1. Push

2. Pop

3. Display

4. Exit

Enter your choice: 3

Stack elements: 10

Stack Operations Menu:

1. Push

2. Pop

3. Display

4. Exit

Enter your choice: 4